

Machine learning evaluation of intensified conditioning on haematopoietic stem cell transplantation in adult acute lymphoblastic leukemia patients

Corresponding Author: Dr Tomoyasu Jo

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This paper is well written and uses machine learning to identify patient conditions where intensified myeloablative conditioning (MAC) is expected to improve outcomes in allogeneic hematopoietic cell transplantation for adult acute lymphoblastic leukemia. Although retrospective, the multi-center, multi-case analysis strengthens the persuasiveness of the results.

1. The reason for limiting the analysis to cases from 2000 to 2021 should be explained. The increasing proportion of older patients over time could have influenced the study's results.
2. Why was overall survival (OS) at one year chosen as the primary endpoint? Is it particularly challenging to identify factors influencing OS beyond one year?
3. How was missing data in the database handled? Given the scale of the dataset, it's reasonable to expect some items to be incomplete.
4. Lastly, there are only 27 unrelated peripheral blood stem cell (UR-PBSC) cases. Is it appropriate to include these data in the analysis, considering their small number?

Reviewer #2

(Remarks to the Author)

This study identifies a subgroup of acute lymphoblastic leukemia (ALL) patients that are more likely to benefit from intensified myeloablative conditioning (MAC) treatment compared to conventional approaches. The analysis is based on a flexible, non-parametric Bayesian machine learning approach (BCF) using regression trees. The results indicate (weakly) significant differences in overall mortality, disease free survival and relapse rate. Subgroup analyses could not identify any individual patient characteristic as having an effect on overall survival.

Overall, this is an interesting study for researchers in the field. Data, analyses and results are presented in a clear way. However, as currently reported, the analysis is not reproducible since important details of the data processing (missingness) as well as of the implemented method (BCF) are not mentioned in text and the code is not available.

Detailed comments:

1. Please report p-values for all stated risk reduction outcomes.
2. line 87: The high risk approach seems non-significant. If included in the abstract, this should be stated clearly.
3. 109: HLA is not defined.
4. 178: What variables had missingness? Was missingness systematic or random? How large was the proportion of missingness that was imputed?
5. 189: The description of the statistical model is unclear and lacks important details.
 - a. Assuming the response variable Y is 1y-mortality, was this coded as binary or continuous?
 - b. 201: When training the model, were default values of bcf R package used for hyperparameters? Please expand.
 - c. 202: What is the rationale to use 1000 regression trees for both BART functions? (The default / recommended settings in the bcf package are 200 for the mean component and 50 for the treatment component, respectively.)
 - d. 204: Please comment on what MCMC diagnostics were carried out. Burn-in of 100 is very low with convergence unlikely at this point. (The authors of bcf recommend "at least a few thousand" in the package documentation.) Equally, 1000 iterations after burn-in is on the low side. Was any thinning used to reduce the potential for correlated posterior draws?

- e. Where multiple MCMC chains run to assess convergence to the posterior distribution?
6. 277: None of the subgroup analyses show a significant effect. Please comment on how this compares to existing literature. Since the observed effect in the main analysis seems to be driven by interaction effects, it would be highly valuable to investigate likely interactions directly, e.g. in additional subgroup analyses. (Although, this might be beyond the scope of this paper.)
7. 285, Fig 2C: The p-values for NRM are essentially the same (and not significant) for both groups. It thus seems unwarranted to state that there is "a trend toward lower NRM" in the high-benefit group.
8. 294, Table 3: These results are difficult to interpret and would benefit from stating associated p-values for the given comparisons. Apart from the high-benefit approach, none of the other approaches seem to show a significant reduction in 1y overall mortality.
9. 317: What is meant by "importance of a comprehensive evaluation of individual patient backgrounds"? Apart from age and sex, were any other demographic, environmental or behavioural covariates included in the model?
10. 346: "These results suggest that those who were transplanted with related grafts are more likely to benefit from intensified MAC." ... This is contradictory to the author's own results in Supp Fig. 4, which shows no significant difference based on graft source. The higher prevalence of related grafts in the high-benefit group should not be interpreted causally.
11. Even if the data cannot be shared, it would be useful to make the analysis code publicly available together with the manuscript.
12. Fig. 2 and Supp. Fig. 2, 5, 6: It would be useful to indicate uncertainty in the shown survival curves, e.g. by adding errorbars or 95% CI error-bands around the curves.
13. Supp Fig. 3: The labels and caption are unclear. Does Q1 correspond to the lowest ITE quartile? Does a negative reduction on the y-axis indicate an increase in deaths?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I have reviewed the revised manuscript and am pleased to see that it has significantly improved. The revisions have effectively addressed all of my previous concerns, and I have no further suggestions or comments.

Reviewer #2

(Remarks to the Author)

I would like to thank the authors for their detailed responses and careful consideration of reviewer comments. I believe the added clarifications and additional analyses make this a very compelling study. I congratulate the authors on their work.

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RESPONSE to Reviewers

Heterogenous Effects of Intensified Conditioning on HSCT for Adult Patients with ALL: A Machine-Learning Approach

Jo T et al.

We thank the reviewers for their thoughtful assessment of the manuscript and their constructive comments. They have significantly strengthened the analysis. Below, we have addressed each comment.

In response to Reviewer #2's comments, we re-ran the model. As a result, the main findings and conclusions were reinforced, though some specific values have changed from the original. Changes to values and related descriptions are highlighted in the manuscript. We have added the references related to the Methods (highlighted).

In addition to the reply to the reviewer's comment, we have carefully checked the manuscript and corrected errors in the manuscript.

“acute lymphoblastic leukemia (ALL)” (page 6, lines 101–102).

“2nd complete remission (CR)” (page 17, lines 336–337).

“1st CR” (page 17, line 338).

Reviewer #1

This paper is well written and uses machine learning to identify patient conditions where intensified myeloablative conditioning (MAC) is expected to improve outcomes in allogeneic hematopoietic cell transplantation for adult acute lymphoblastic leukemia. Although retrospective, the multi-center, multi-case analysis strengthens the persuasiveness of the results.

> Response: Thank you for your deep understanding and your positive comments.

1. The reason for limiting the analysis to cases from 2000 to 2021 should be explained. The increasing proportion of older patients over time could have influenced the study's results.

> Response: Thank you for your valuable comment. To develop a robust machine-learning model, a large dataset was necessary, which led us to select a relatively broad

range for the transplantation period. Since cases transplanted in early periods tend to have a higher rate of missing data, we limited our analysis to cases from the year 2000 onward, where the missing data were less prevalent. As you pointed out, the proportion of older patients has been increasing over time. In this study, we included both the transplantation period and the patients' age as factors in the model development, thereby adjusting the results for these variables. In the high-benefit group, which is more likely to benefit from the intensified myeloablative conditioning (MAC), the proportion of transplantation performed in the earlier period was higher, suggesting that the survival benefit provided by the intensified MAC may have become relatively smaller over time, possibly due to advancements in transplantation procedures, as well as changes in patient demographics. We have added a description on this point in the Discussion part of the revised manuscript.

“Fourth, although the subgroup analysis in this study did not show a difference in the benefit of intensified MAC across different transplantation periods, the high-benefit group had a higher proportion of cases with transplants conducted in earlier periods. The potential impact of ongoing trends, such as the increase in older patients, and advancements in transplant procedures, will require further evaluation in the future.” (page 20, lines 402–407).

2. Why was overall survival (OS) at one year chosen as the primary endpoint? Is it particularly challenging to identify factors influencing OS beyond one year?

> Response: Thank you for your valuable comment. In principle, it is possible to set an endpoint for overall survival (OS) longer than one year using the method employed in this study. We selected 1-year OS as the endpoint for two reasons. First, since this study focuses on the intensity of the conditioning, we chose a relatively short-term endpoint, as the effect of conditioning intensity are likely to be more apparent within one year. Second, it relates to case exclusion. In the current analysis, where 1-year OS was used as the endpoint, we excluded cases that were alive at the last follow-up, but had a follow-up period of less than one year from the analysis cohort. If we set a longer OS endpoint, such as 3-year OS, for survivors, only those who were followed up for more than three years would be included, excluding survivors with follow-up periods shorter than three years. However, for patients who died within three years after transplantation, all would be included. This would increase the proportion of survivors excluded from the analysis due to length of follow-up period, potentially impacting the results. Although a large proportion of cases in this study had a follow-up period longer than one year, we cannot rule out the impact of case exclusions on the results. We have added this point to the

limitations in the revised manuscript.

“Second, we focused on reduction in 1-year overall mortality, as the effects of conditioning intensity are likely to be more apparent within this timeframe. Additionally, setting the endpoint at one year helped minimize exclusion of cases due to shorter observation period. However, further analysis regarding longer-term prognosis is still necessary.” (page 19, lines 392–396).

3. How was missing data in the database handled? Given the scale of the dataset, it’s reasonable to expect some items to be incomplete.

> Response: Thank you for your important comment. We described the missing value for each variable among the entire cohort before propensity score matching in the Supplemental Table 1. For example, CMV antibody positivity was missing in 8.4% of cases in the standard myeloablative conditioning (MAC) group, and 4.6% in the intensified MAC group. These missingness was imputed using a random forest algorithm, then the analysis was conducted on the cohort after propensity score matching. While missing data were minimal for variables critically related to relapse, and non-relapse mortality, such as age, sex, disease phenotype, refined Disease Risk Index (rDRI), graft source, and HLA mismatch, there is still a potential for these missing values to have influenced the results. Therefore, we have added this point as a limitation in the revised manuscript.

“Some of the factors included in the analysis contained missing values. Although we imputed them with random forest imputation approach, we cannot rule out the potential bias due to missingness.” (pages 19–20, lines 400–402).

4. Lastly, there are only 27 unrelated peripheral blood stem cell (UR-PBSC) cases. Is it appropriate to include these data in the analysis, considering their small number?

> Response: Thank you for your important comment. In this study, the high-benefit group, which is more likely to benefit from intensified myeloablative conditioning (MAC), showed a tendency to have fewer unrelated donor bone marrow and unrelated donor PBSC cases. As you pointed out, because the number of unrelated PBSC cases was small, it is difficult to evaluate the detailed impact of unrelated donor PBSC on the benefit of intensified MAC. However, since we have incorporated five graft sources as variables: related bone marrow, related PBSC, unrelated bone marrow, unrelated PBSC, and cord blood in the development of the machine-learning model, we believe that including UR-

PBSC cases does not bias the overall results. Moreover, since 2010, when the Japan Marrow Donor Program started facilitating UR-PBSC transplants, the number of unrelated donor peripheral blood stem cell (UR-PBSC) transplants has gradually increased in Japan, and this upward trend is expected to continue. Based on this, UR-PBSC cases are included in this study. We have noted this point in the revised manuscript.

“While, in Japan, the number of transplants using unrelated PBSCs has gradually increased since 2010, when the Japan Marrow Donor Program began facilitating unrelated PBSC, the number of unrelated PBSC cases remained small in the current analysis. Further analysis is needed to evaluate impact of selecting unrelated PBSCs on the benefit of intensified MAC.” (page 18, lines 358–362).

Reviewer #2

This study identifies a subgroup of acute lymphoblastic leukemia (ALL) patients that are more likely to benefit from intensified myeloablative conditioning (MAC) treatment compared to conventional approaches. The analysis is based on a flexible, non-parametric Bayesian machine learning approach (BCF) using regression trees. The results indicate (weakly) significant differences in overall mortality, disease free survival and relapse rate. Subgroup analyses could not identify any individual patient characteristic as having an effect on overall survival. Overall, this is an interesting study for researchers in the field. Data, analyses and results are presented in a clear way.

> Response: Thank you for your deep understanding and your positive comments.

However, as currently reported, the analysis is not reproducible since important details of the data processing (missingness) as well as of the implemented method (BCF) are not mentioned in text and the code is not available.

> Response: We appreciate your helpful suggestions. We clarified the data processing and shared the R codes in the Github. Please see our response to each of your comments for details.

Detailed comments:

1. Please report p-values for all stated risk reduction outcomes.

> Response: Thank you for your important comment. We have added p-values in the revised manuscript (page 4, lines 87–90).

2. line 87: The high risk approach seems non-significant. If included in the abstract, this should be stated clearly.

> Response: Thank you for your important comment. We have revised the manuscript so as to state clearly that reduction in 1-year overall mortality with the high-risk approach was not significant, compared to the population approach.

“In contrast, the high-risk approach (targeting patients with high or very high rDRI) did not achieve statistical significance (risk difference [95% confidence interval], +3.85 percentage points [-1.11 to 7.90], $p = 0.063$).” (page 4, lines 88–90).

3. 109: HLA is not defined.

> Response: Thank you for your helpful comment. We have clarified that HLA stands for

“human leukocyte antigen” in the revised manuscript (page 6, lines 111–112).

4. 178: What variables had missingness? Was missingness systematic or random? How large was the proportion of missingness that was imputed?

> Response: Thank you for your important comment. We described the missing value for each variable among the entire cohort before propensity score matching in the Supplemental Table 1. For example, CMV antibody positivity was missing in 8.4% of cases in the standard myeloablative conditioning (MAC) group, and 4.6% in the intensified MAC group. The distribution of missing data did not show clear difference between the intensified MAC group and the standard MAC group, suggesting a random distribution. All the missing values were imputed using a random forest algorithm, then the analysis was conducted on the cohort after propensity score matching. While missing data were minimal for variables critically related to relapse, and non-relapse mortality, such as age, sex, disease phenotype, refined Disease Risk Index (rDRI), graft source, and HLA mismatch, there is still a potential for these missing values to have influenced the results. Therefore, we have added this point as a limitation in the revised manuscript.

“Some of the factors included in the analysis contained missing values. Although we imputed them with random forest imputation approach, we cannot rule out the potential bias due to missingness.” (pages 19–20, lines 400–402).

5. 189: The description of the statistical model is unclear and lacks important details.

a. Assuming the response variable Y is 1y-mortality, was this coded as binary or continuous?

b. 201: When training the model, were default values of bcf R package used for hyperparameters? Please expand.

c. 202: What is the rationale to use 1000 regression trees for both BART functions? (The default / recommended settings in the bcf package are 200 for the mean component and 50 for the treatment component, respectively.)

d. 204: Please comment on what MCMC diagnostics were carried out. Burn-in of 100 is very low with convergence unlikely at this point. (The authors of bcf recommend "at least a few thousand" in the package documentation.) Equally, 1000 iterations after burn-in is on the low side. Was any thinning used to reduce the potential for correlated posterior draws?

e. Where multiple MCMC chains run to assess convergence to the posterior distribution?

> Response: We appreciate your insightful comments on statistical model.

- a. The response variable was binary variable (i.e., 1-year mortality), and thus the estimand of our model was risk difference.
- b. We applied default values when running *bcf* package in R otherwise clarified.
- c. In the revised manuscript, we re-ran the model with 200 trees for the mean component and 50 trees for the treatment component, as defined by the default.
- d. In the revised manuscript, we re-ran the model with 2500 burn-in MCMC iterations and 2500 MCMC iterations to save after burn-in. The results did not qualitatively change from our original submission.
- e. We calculated the potential scale reduction factor (PSRF) in Gelman and Rubin's MCMC Convergence Diagnostic, and confirmed that MCMC was converged (PSRF=1.08).

We clarified these points in the revised manuscript.

“For the purpose of this study, we used 1-year overall mortality after HSCT (binary outcome).” (page 8, lines 160–161).

“First, after describing baseline characteristics before and after propensity score matching, we applied a machine learning BCF algorithm (*bcf* package in R)” (page 10, line 193).

“In our analysis, we built BCF model by constructing 200 and 50 regression trees including all covariates mentioned above for the prognostic function and treatment effects, respectively. The model was trained through 2500 burn-in and 2500 Markov chain Monte Carlo iterations.” (page 10, lines 201–205).

“The convergence of Markov chain Monte Carlo (MCMC) algorithm was assessed by calculating the potential scale reduction factor (PSRF) in Gelman and Rubin's MCMC Convergence Diagnostic.” (pages 10, lines 205–208).

“PSRF was 1.08, indicating the convergence of MCMC algorithm.” (page 14, line 262).

6. 277: None of the subgroup analyses show a significant effect. Please comment on how this compares to existing literature.

> Response: Thank you for your important comment. In the previous studies on intensified myeloablative conditioning (MAC) for adult acute lymphoblastic leukemia (ALL), the benefits of intensified MAC were reported in the Philadelphia (Ph)

chromosome positive subgroup (Morita-Fujita M, et al. Hematol Oncol. 2022;40:1041; Arai Y, et al. Am J Hematol. 2018;93:47) or in patients who underwent cord blood (CB) transplantation (Arai Y, et al. Bone Marrow Transplant. 2016;51:1636). In the current analysis, we also added the results of a subgroup analysis based on the presence of Ph chromosome to the revised Supplemental Figure 4. In our study, there was a tendency for the Ph-positive subgroup to derive benefit from intensified MAC compared to Ph-negative subgroup, though this was not statistically significant. While Ph chromosome has historically been considered a poor prognostic factor, recent advancement in tyrosine kinase inhibitors have led to significant progress in pre- and post-transplant treatments compared to previous era, that may have impacted outcomes. For CB transplantation, as shown in the Supplemental Figure 4, no significant improvement in overall survival (OS) was observed from intensified MAC in the CB transplantation subgroup in our study. The present analysis included patients who were not in remission at the time of transplantation (while previous reports were limited to patient in complete remission at transplantation), suggesting that differences in patient backgrounds may have influenced the results. We have added a discussion on how the current results correspond with previous literature in the Discussion section in the revised manuscript.

“In the previous studies on intensified MAC for ALL patients, the benefits of intensified MAC were reported in the Ph chromosome positive subgroup, or in patients who underwent CB transplantation. In our study, there was a tendency for the Ph-positive subgroup to derive benefit from intensified MAC compared to Ph-negative subgroup, though this was not statistically significant. While Ph chromosome has traditionally been considered a poor prognostic factor, recent advancement in tyrosine kinase inhibitors have led to significant progress in pre- and post-transplant treatments, that may have impacted outcomes. For CB transplantation, no significant improvement in OS was observed from intensified MAC in the CB transplantation subgroup in our study. The present analysis included patients who were not in remission at the time of transplantation (while previous reports were limited to patient in complete remission at transplantation), suggesting that differences in patient backgrounds may have influenced the results.” (page 18, lines 364–375).

Since the observed effect in the main analysis seems to be driven by interaction effects, it would be highly valuable to investigate likely interactions directly, e.g. in additional subgroup analyses. (Although, this might be beyond the scope of this paper.)

> Response: Thank you for your valuable comment. In the subgroup analysis, none of the

individual variables could identify patient group that would benefit from the intensified myeloablative conditioning (MAC). Therefore, as you suggested, we performed an additional subgroup analysis using a combination of three variables; age ($<$ median, or $\geq$ median), refined Disease Risk Index (rDRI) (intermediate, or high/very high), and graft source (related bone marrow/peripheral blood stem cell [PBSC], unrelated bone marrow/PBSC, or unrelated cord blood [CB]) (Supplemental Figure 7). While younger patients tended to benefit more from the intensified MAC compared to older patients, and patients with intermediate rDRI tended to benefit from the intensified MAC compared to those with high/very high rDRI, it was still difficult to identify patient group that would benefit from intensified MAC using these three factors. In particular, impact of graft source interacted with age and rDRI. For example, among younger patients with intermediate rDRI, those transplanted with unrelated CB tended to benefit from the intensified MAC. However, among younger patients with high/very high rDRI, and older patients, benefit of the intensified MAC was smaller in those who received unrelated CB transplantation, suggesting presence of interaction. These findings indicate that identifying groups that would benefit from the intensified MAC based on a few variables is challenging, underscoring the usefulness of machine learning-based approach, which can evaluate multiple variables. We have added the Supplemental Figure 7 and related description to the Results part and the Discussion part in the revised manuscript.

“we found no evidence of heterogeneity in the subgroup analysis by each characteristic (Supplemental Figures 4–6), or by combination of several characteristics (Supplemental Figure 7).” (page 14, lines 279–281).

“determined by a combination of multiple patient background factors rather than a single variable or a set of a few variables.” (page 15, lines 291–292).

“they could not identify patients who benefit from intensified conditioning, as a single variable, nor could they do so when combined with a few factors (Supplemental Figures 4, and 7).” (pages 18–19, lines 377–379).

7. 285, Fig 2C: The p-values for NRM are essentially the same (and not significant) for both groups. It thus seems unwarranted to state that there is "a trend toward lower NRM" in the high-benefit group.

> Response: Thank you for your important comment. As you pointed out, the trends for better overall survival (OS) and a significantly better disease-free survival (DFS) in the

high-benefit group were due to a trend for reduced relapse without an increase in non-relapse mortality. We have revised the description. And, following your suggestion in Comment #5 above, we have re-run the model, which has resulted in changes to the values described in Figure 2. These updates have been incorporated into the revised manuscript.

“we found a significantly better DFS (HR, 0.839; 95% CI, 0.717–0.983; $p = 0.030$) with intensified conditioning in the high-benefit group, but not in the low benefit group (Figure 2). Although we found similar trends for OS (HR, 0.877; 95% CI, 0.741–1.037; $p = 0.124$) and for a reduction in relapse (HR, 0.847; 95% CI, 0.692–1.037; $p = 0.107$), the 95% CIs included the null values. We did not find an increase in NRM with intensified conditioning in the high-benefit group. In the low-benefit group, we found no improvement in OS, DFS, or relapse, and no increase in NRM with intensified conditioning.” (pages 14–15, lines 283–290).

8. 294, Table 3: These results are difficult to interpret and would benefit from stating associated p-values for the given comparisons. Apart from the high-benefit approach, none of the other approaches seem to show a significant reduction in 1y overall mortality.

> Response: Thank you for your important comment. As you pointed out, in approaches other than the high-benefit approach, we did not observe a significant reduction in 1-year overall mortality with intensified myeloablative conditioning (MAC). We have included p-values in the revised Table 3, and have revised related description in the manuscript. Please note that we have re-run the model following your suggestion in Comment #5 above, but the results did not qualitatively change. These updates have been incorporated into the revised manuscript.

“we found the strongest association in the high-benefit approach with a reduction in 1-year overall mortality (+5.93 percentage points [95% CI, 0.88 to 10.51]). In contrast, reductions observed with the high-risk approach (+3.84 percentage points [95% CI, -1.11 to 7.90]), the modified population approach (+2.87 percentage points [95% CI, -3.16 to 9.03]), and the population approach (reduction +2.05 percentage points [95% CI, -1.14 to 5.26]) were not significant (Table 3).” (page 15, lines 296–302).

9. 317: What is meant by "importance of a comprehensive evaluation of individual patient backgrounds"? Apart from age and sex, were any other demographic, environmental or behavioral covariates included in the model?

> Response: Thank you for your important comment. In the model, we included the factors described in Table 1, but we did not include environmental or behavioral factors. As you pointed out, the term “comprehensive evaluation” may be unclear, so we have revised the description.

“The heterogenous ITEs observed in this study highlight the importance of comprehensively considering and evaluating various clinical factors related to patient or disease characteristic, as well as transplantation procedures, when determining which patients should receive intensified myeloablative conditioning” (page 16, lines 320–323).

10. 346: "These results suggest that those who were transplanted with related grafts are more likely to benefit from intensified MAC." ... This is contradictory to the author's own results in Supp Fig. 4, which shows no significant difference based on graft source. The higher prevalence of related grafts in the high-benefit group should not be interpreted causally.

> Response: Thank you for your valuable comment. Table 2 shows that in the high-benefit group, which is more likely to benefit from the intensified myeloablative conditioning (MAC), the proportion of related donor transplants was significantly higher compared to the low-benefit group. However, in the subgroup analysis shown in the Supplemental Figure 4, although there was a trend favoring the intensified MAC in the subgroup that received related donor transplants, it did not reach statistical significance. Therefore, while the results of this study suggest a potential relationship between graft source and the benefit of the intensified MAC, it is likely that other factors also have a significant influence. We have revised the description in the manuscript accordingly.

“These results suggest that the benefit of the intensified MAC may potentially differ depending on graft source” (page 17, lines 351–352).

11. Even if the data cannot be shared, it would be useful to make the analysis code publicly available together with the manuscript.

> Response: As suggested, we shared the R codes for our BCF model in the Github page (https://github.com/Koinoue/bcf_hsct). We also clarified this in the revised manuscript (page 3, lines 52–53).

12. Fig. 2 and Supp. Fig. 2, 5, 6: It would be useful to indicate uncertainty in the shown survival curves, e.g. by adding error bars or 95% CI error-bands around the curves.

> Response: Thank you for your helpful comment. We have added 95% confidence interval error bands around curves to the revised Figure 2 and Supplemental Figures 2, 5, and 6, and related legends for each figure.

13. Supp Fig. 3: The labels and caption are unclear. Does Q1 correspond to the lowest ITE quartile? Does a negative reduction on the y-axis indicate an increase in deaths?

> Response: Thank you for your helpful comment. Please note that we have re-run the model following your suggestion in Comment #5 above, but the results did not qualitatively change. These updates have been incorporated into the revised manuscript. Throughout the manuscript, the causal estimand was defined as the reduction in 1-year death by intensified MAC (vs. non-intensified MAC). Thus, Q1 represents the group with the lowest ITEs. Y-axis shows “the reduction in 1-year death”. In contrast, in Q3 and Q4, where ITEs are higher than the median value, 1-year mortality decreases with the intensified MAC. We have added these explanations to the legend of Supplemental Figure 3 as well as the manuscript.

“Second, based on the BCF model, we predicted the ITE (i.e., the reduction in 1-year mortality by intensified MAC compared to non-intensified MAC) for each patient.” (page 11, lines 212–213).

RESPONSE to Reviewers

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We thank the reviewers for their thoughtful assessment of the manuscript and their constructive comments.

Reviewer #1

I have reviewed the revised manuscript and am pleased to see that it has significantly improved. The revisions have effectively addressed all of my previous concerns, and I have no further suggestions or comments.

> Response: We are pleased to hear that the revisions have significantly improved the manuscript and that all previous concerns have been addressed satisfactory.

Reviewer #2

I would like to thank the authors for their detailed responses and careful consideration of reviewer comments. I believe the added clarifications and additional analyses make this a very compelling study. I congratulate the authors on their work.

> Response: We are delighted that the reviewer finds our study compelling and appreciate their congratulations on our work.